Cleveland, Ohio

A strong El Niño usually gives Cleveland a milder winter, and 76 winters of station data confirm it. That is the danger. Winter deaths here come from single cold nights, heating fires, and gas shutoffs, and none of those needs a harsh season. This winter the federal heating money is also on a stopgap, so the threat is a cash gap more than a weather one.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
City of Cleveland, about 363,600 people, inside Cuyahoga County (several Climate Impact Lab regions)	No US figure published; the projection covers heat only	Peak risk December 2026 to February 2027	Sudden cold, heating fires and carbon monoxide, flash-freeze crashes, and disconnection

17/17 recommendations verified. 1 confirmed, 8 corrected (partially confirmed), 1 unverified, 7 contraindicated. The funding target moved from $1.5M to $1.325M, down 12%, so the envelope stands. **The El Niño premise was tested, not assumed:** a verifier pulled every Cleveland winter since 1950 and classed each one by the official El Niño index. **The run's biggest correction:** the Philanthropist's 60 extra shelter beds would have double-funded overflow the county levy already pays for, so that line was halved. **Resolution caveat:** a city, not a ward. Actions name the east-side neighbourhoods the lenses found most exposed, but no ward-level data were verified. **Method caveat:** web search was exhausted; agents worked by direct fetch, and several state pages would not load.

EXECUTIVE SUMMARY — AT A GLANCE



41.50°N 81.69°W, United States of America · Natural Earth

- **The risk.** A strong El Niño usually means a milder winter, but deaths here come from single cold nights, heating fires, and gas shutoffs, and this winter the federal heating money is on a stopgap.
- **First action.** Ohio bridges the federal heating cash gap, and the City of Cleveland pauses city utility shutoffs if LIHEAP has not arrived by 1 November.
- **The money.** $575k floor, $1.325M target, $2.025M ceiling ($2.925M if Winter Crisis goes unfunded); the money covers the cold nights and cash gap that state and federal programs leave.
- **Do not.** Do not hand out space heaters, and do not promote generators: improvised heat is the fire and carbon monoxide pathway.*

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* Suggested formulation: check with a health worker before sharing.

Cleveland Hopkins station record, winters 1950–51 to 2025–26, classed by the December–February Oceanic Niño Index. Lake Erie ice is the annual maximum, recorded from 1973.

WINTERS	SNOW (IN)	DEC–FEB MEAN (°F)	DAYS AT OR BELOW 0 °F	WINTERS WITH ANY 0 °F DAY	LAKE ERIE MAX ICE
Strong El Niño (8)	44.5	33.0	0.9	3 of 8	63%
All winters (76)	57.8	29.4	4.1	58 of 76	83% (other years)
Moderate or weak El Niño (18)	61.4	28.1	4.7	16 of 18	—
La Niña (26)	59.6	29.8	3.3	19 of 26	—

- **On average, strong El Niño lowers the hazard.** Winters run about 3.6 °F warmer with about 23% less snow, and days at or below 0 °F fall from about four a winter to about one.
- **It is not reliable year to year.** 1972–73 and 1991–92 were snowier than average. Three of the eight strong winters had sub-zero days, and 1957–58 reached −6 °F. The strong-autumn winters of 1965–66 and 1987–88 reached −6 °F and −5 °F.
- **The quiet recent winters are partly the warming trend.** None of the four strong El Niños since 1997 had a sub-zero day, but winters since 1991–92 also average fewer sub-zero days overall. El Niño cannot take all the credit.
- **Low ice keeps lake-effect snow going.** Lake Erie iced to only 5.4% in 1997–98, the lowest on record. Open water feeds lake-effect bursts into January and February, which fall hardest on the east side.
- **Winter deaths barely track how cold the winter is.** Across 36 US cities, excess winter mortality "was no lower in warmer vs. colder cities" (Kinney et al. 2015). Influenza, heart strain, and cold homes drive it. A mild forecast therefore buys little protection.

- **Poverty.** About a third of residents live below the poverty line, and median household income is about $30,900 (American Community Survey via Wikipedia; not independently verified).

* Suggested formulation: check with a health worker before sharing.

- **Heating costs.** Low-income households in Cleveland spend a median 9.1% of income on energy, and one in four spends 15.7% or more; the national low-income median is 8.3% (ACEEE, 2019 housing survey data). A 10 Mcf Enbridge gas bill rose from $69 in February 2020 to $148 in February 2026. NEADA projects the average US heating bill at $1,030 this winter, up 8.7%.
- **Federal money is on a stopgap.** Federal staff running the Low Income Home Energy Assistance Program (LIHEAP) were fired on 1 April 2025. The program was proposed for elimination in both FY26 and FY27. FY26 was funded, with $181.2M for Ohio, but the first release arrived on 28 November 2025. A continuing resolution signed 2 September 2026 funds it at last year's level only to 11 December, and no October release has been announced.
- **Unsheltered people.** The city's rapid-rehousing push cut unsheltered numbers from about 300 to about 100. Separately, new HUD rules put about 400 county-funded *housing* beds at risk; these are not emergency shelter beds.
- **Submetered renters** cannot enrol in Ohio's percentage-of-income payment plan (PIPP). The governor vetoed HB 173 on 25 June 2026, preserving their court-won protections, but the enrolment gap remains.
- **Carbon monoxide.** The Cuyahoga coroner recorded 122 carbon monoxide deaths in 2000–03; 69% were accidental, and they clustered among Cleveland residents. No study links Ohio shutoffs to carbon monoxide deaths.

WHAT THIS PLACE SHOULD DO

RANKED BY LIVES PER EFFORT

The panel converged on one finding from four directions: a mild El Niño winter is the one where agencies stand down, and people die on the night it turns. The first three actions keep heat on and people housed through that night. The rest close the specific ways Cleveland's past winters have killed.

Keep the heat on: bridge the federal cash gap, and pause city utility shutoffs if LIHEAP has not arrived by 1 November

PARTIALLY CONFIRMED | ECONOMIST · PUBLIC HEALTH · PHILANTHROPIST · FRONTLINE | ADAPTATION INVENTORY: PRECEDENT, US CONTINGENT LOANS

OWNER	TRIGGER	COST	PREVENTS
Ohio (HEAP, now run by Job and Family Services); City of Cleveland for Cleveland Public Power and Cleveland Water	No federal LIHEAP release to Ohio by 1 November	State: interest on an advance, about $1–2M (estimate). City: carrying deferred bills, about $0.1–0.3M (estimate)	Disconnection in the November–December cash gap, and the improvised heating that follows

* Suggested formulation: check with a health worker before sharing.

The Economist's one thing: the uninsured shock is timing, not weather. Last winter's first federal release came on 28 November, and a disconnection is really a loan the household cannot get. A line of credit repaid from the federal award costs about 1% of a grant covering the same gap. The city ran a 30-day moratorium at its own utilities from 7 November 2025, so the precedent exists. Falling heating prices averted about 12,500 US deaths a year, mostly cardiovascular and respiratory (Chirakijja, Jayachandran and Ong, national evidence). Whether Ohio can legally advance HEAP money was not verified, and no source confirms that Cleveland Public Power follows the state's winter reconnect order voluntarily.

October door-knock: check documents and enrol households in HEAP and PIPP through Step Forward now, while the gas is still on

PARTIALLY CONFIRMED · FRONTLINE · EMERGENCY MGR · HISTORIAN · ECONOMIST · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Step Forward (the only Cuyahoga HEAP and PIPP agency); block volunteers; NEOCH's ID collaborative	Through October	About $150k for canvass staff (estimate); volunteers otherwise	Winter Crisis applications stalling in December for missing papers

The Frontline lens's one thing: nobody calls until the gas is already off. The Winter Crisis Program opens 1 November and pays only after a disconnect notice, for households at or below 175% of the poverty line. Regular HEAP and PIPP cost nothing, and PIPP caps the bill at 5–10% of income, but processing can take up to 12 weeks, so an application now barely lands before January. Use the neighbourhood

* Suggested formulation: check with a health worker before sharing.

anchors people already trust, hold sign-up nights at the Housing Court kiosk branches, and send anyone without ID to NEOCH.
Corrections: CHN Housing Partners no longer processes HEAP; apply through Step Forward (216-480-4327) or the state's Job and Family Services portal, not the old development.ohio.gov pages.

Write a Code Blue trigger now, pre-release overflow and street outreach on it, and ring-fence the January–February budget

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · PHILANTHROPIST · PUBLIC HEALTH · ECONOMIST

ADAPTATION INVENTORY: PRECEDENT, US WINTER WEATHER ADVISORIES

OWNER	TRIGGER	COST	PREVENTS
Cuyahoga Office of Homeless Services and Cleveland Public Health; NEOCH and Lutheran Metropolitan Ministry; city Parks and Recreation	Wind chill at or below 0 °F, a freezing-rain warning, or a 30 °F drop within 12 hours	$200k grant; warming centres about $15–25k per activation-week (estimate)	Exposure deaths among the roughly 100 unsheltered people

* Suggested formulation: check with a health worker before sharing.

The Emergency Manager's one thing: by mid-January in a warm winter everyone has stood down, and then one Arctic front drops the temperature 40 degrees overnight. Write the trigger on the 3–5 day forecast, not the seasonal outlook. Last January four recreation centres opened and drew more than 430 visitors. Lutheran Metropolitan Ministry's shelter holds up to 365 men with 30–60 partner overflow beds, already levy-funded, so the grant pays for *extra* cold-night beds and one outreach team only. Warming centres alone fail when people cannot reach them: pair them with outreach vans and phone lists.

Install carbon monoxide and smoke alarms in households that have had a shutoff notice, with a "no ovens, no generators indoors" message

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · SKEPTIC · PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Cleveland Division of Fire with the Red Cross; Step Forward to supply addresses	Alongside the October canvass	$125k, about 5,000 CO alarms at $25 each (device only)	Heating fires and carbon monoxide deaths from improvised heat

In Winter Storm Elliott, Ohio lost a family of six to a house fire from unsafe heating; the fire's location was not given. About 26% of low-income US households used risky heating such as ovens or space heaters (Carley et al. 2022, survey). Cleveland Fire already installs free smoke alarms through the Red Cross, so this money buys carbon monoxide alarms only. Whether the Red Cross program fits CO alarms was not confirmed, and furnace safety checks need their own budget.*

Agree a flash-freeze road-closure trigger before the crash, not after it

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · SKEPTIC | ADAPTATION INVENTORY: PRECEDENT, US WINTER WEATHER ADVISORIES

OWNER	BY WHEN	COST	PREVENTS
ODOT District 12, Ohio Turnpike Commission, NWS Cleveland	1 December	Near zero	Pileups like Elliott's, where 8 of Ohio's 16 deaths were in vehicles

* Suggested formulation: check with a health worker before sharing.

Elliott dropped only 3.6 inches of snow on Cleveland, yet four people died in a roughly 50-vehicle Turnpike pileup in Sandusky County. The 1978 blizzard closed the entire Turnpike for the first time. In Buffalo in 2022, two-thirds of rescue vehicles sent out needed rescuing themselves. Pre-position four-wheel-drive and ambulance capacity before any driving ban. No evaluation of pre-emptive closure triggers was found.

Flu and COVID vaccination for adults over 65 and heart patients, with cold-snap check calls to heart and COPD patients

PARTIALLY CONFIRMED | PUBLIC HEALTH | ADAPTATION INVENTORY: PRECEDENT, US COMMUNITY HEALTH CENTRES

OWNER	BY WHEN	COST	PREVENTS
Cleveland Public Health; health systems and community health centres	Vaccination by November; calls on each trigger	Existing budgets	The cardiorespiratory deaths that make up most winter excess mortality

The Public Health lens's one thing: a mild winter will probably not lower Cleveland's winter deaths much, because they are driven by respiratory viruses and heart strain rather than the cold itself. No Cleveland trial of this action was found.

Get submetered tenants into PIPP through a regulator filing

CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Ohio Consumers' Counsel or PUCO	File now	Near zero; the existing PIPP rider pays	The one group with no bill protection at all staying unprotected

* Suggested formulation: check with a health worker before sharing.

The governor's own veto statement says submetered customers "can't enroll" in PIPP. He was speaking about electricity. A rule change will not land before this peak, but the filing costs nothing and protects every winter after.

Hold a bridge reserve for the housing beds at risk under the new HUD rules

PARTIALLY CONFIRMED · PHILANTHROPIST · ADAPTATION INVENTORY: NONE FOUND

OWNER	TRIGGER	COST	PREVENTS
Cuyahoga Continuum of Care lead agency	HUD award results show lost renewals	$0.5–1M, held until awards are known	People losing housing mid-winter if renewals are cut

HUD now guarantees only 60% of renewal funding, down from 90–96%. Applications were due 26 August and county contracts would start 1 November; the award date is unknown. The roughly 400 beds are probably housing beds, not shelter beds.

After the peak: weatherise the homes of households that needed Winter Crisis help

PARTIALLY CONFIRMED · AFTER THE PEAK · PUBLIC HEALTH · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Step Forward's weatherisation program (not the city ARPA home-repair channel)	Spring 2027 onward	Existing weatherisation budgets	Cold homes and respiratory illness in later winters

* Suggested formulation: check with a health worker before sharing.

A New Zealand trial found wheezing fell (odds ratio 0.57) and GP visits fell (0.73); hospital admissions showed only a non-significant trend. A 2024 review found no evidence either way on mortality. The right investment, but not for this window.

Summer 2027: confirm the water utility can test for algal toxins and has backup treatment ready

UNVERIFIED HISTORIAN ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Cleveland Water	Before July 2027	Existing budget	A repeat of Toledo's 2014 do-not-drink order

In August 2014 an algal bloom over Toledo's intake left more than half a million people without drinkable tap water for three days. The claim that a mild, wet El Niño winter raises spring runoff into Lake Erie was not verified; this is a prudent check, not a forecast.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Ohio Job & Family Services (HEAP)	City of Cleveland (CPP, Water, Parks)	Step Forward	Block volunteers	NEOCH	Lutheran Metropolitan Ministry	Cuyahoga Office of Homeless Services	Cleveland Public Health	Cleveland Fire & Red Cross	ODOT District 12 & Ohio Turnpike	NWS Cleveland	Health systems & health centres	Ohio Consumers' Counsel & PUCO	Cuyahoga CoC lead agency
1. Bridge the LIHEAP gap; pause city shutoffs	**A/R**	**R**	**I**											
2. October door-knock: HEAP and PIPP enrolment	**I**		**A/R**	**R**	**R**									
3. Code Blue trigger, overflow, and outreach		**R**			**R**	**R**	**A**	**R**			**C**			
4. CO and smoke alarms after shutoff notices			**R**						**A/R**					
5. Flash-freeze road-closure trigger										**A/R**	**R**			
6. Vaccination and cold-snap check calls								**A**				**R**		
7. Submetered tenants into PIPP													**A/R**	
8. Bridge reserve for at-risk housing beds														**A/R**
9. Weatherise Winter Crisis households			**A/R**											
10. Algal-toxin testing and backup treatment		**A/R**												

What the chart shows. **Step Forward appears on four rows and is accountable for two**. It is the only HEAP and PIPP agency in Cuyahoga, supplies the addresses for the alarm row, and would hold the $500k arrears fund and the $0.9M reconnect contingency, so it is the plan's single point of failure. **Row 1 has two public owners and an unverified legal basis**: the state runs HEAP and the city runs its own utilities, and whether Ohio can legally advance HEAP money was not verified. **Block volunteers carry an R on the door-knock**, the action that decides whether Winter Crisis applications land in time. **Rows 7 to 10 each rest on a single body**, and none of them lands before this winter's peak.

WHERE MONEY DOES THE MOST

FUNDING

Verified and corrected. The money is small beside the state and federal heating programs; its job is to cover the cold nights and the cash gap those programs leave, through organisations that can spend within weeks.

FLOOR

TARGET

CEILING

LARGEST UNCERTAINTY

* Suggested formulation: check with a health worker before sharing.

$575k	$1.325M	$2.025M, or $2.925M if Winter Crisis goes unfunded	Whether HEAP Winter Crisis is funded for 2026–27

Arithmetic. Floor: cold-night overflow and outreach $200k (30 beds × 60 nights × $75 = $135k, plus one team × 16 weeks × $4k = $64k) + arrears at half size $250k + CO alarms $125k = $575k. Target: overflow $200k + arrears $500k (1,000 households × $500) + CO alarms $125k + housing bridge $500k = $1.325M. Ceiling: overflow $400k + arrears $500k + alarms $125k + bridge $1M = $2.025M. A $0.9M reconnect fund (5,000 households × $175) is held as a contingency, not a base line, because it duplicates Winter Crisis while that program is funded. The $75 bed-night and $500 arrears figures are assumptions with no published source. The target fell 12% from the lens's $1.5M.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
NEOCH and Lutheran Metropolitan Ministry	Code Blue extra beds and one outreach team	$200k	Restricted grant released on the temperature trigger	Highest concentrated risk; pipeline already delivers	Keep, halved
Step Forward	Arrears for households just over the HEAP limit or past their Winter Crisis cap	$500k	Pooled fund paid direct to the utility	Stops shutoffs that lead to improvised heat	Keep, conditional
Cleveland Fire via Step Forward intake	Carbon monoxide alarms	$125k	Bulk purchase and install	Device cost only; smoke alarms already free	Keep, re-scoped
Continuum of Care lead agency	Housing bridge reserve	$0.5–1M	Held until HUD awards	60% renewal guarantee	Keep, hold
Step Forward	$175 reconnect payments	$0.9M	Contingency fund	Only if Winter Crisis goes unfunded	Contingency

* Suggested formulation: check with a health worker before sharing.

DO NOT DO THIS CONTRAINDICATED

Do not read "strong El Niño, mild winter" as a reason to stand down

CONTRAINDICATED SKEPTIC · EMERGENCY MGR · HISTORIAN · PUBLIC HEALTH; CONFIRMED BY STATION DATA

The average is milder, but three of eight strong El Niño winters still had sub-zero days, and winter mortality barely tracks winter severity. Buffalo's 41 deaths in 2022 came from one three-day storm, not a harsh season.

Instead: trigger on 3–5 day forecasts and keep the January–February budget ring-fenced.

Do not hand out space heaters, and do not promote generators*

CONTRAINDICATED SKEPTIC; CONFIRMED WITH CORRECTIONS

Improvised heat is the fire and carbon monoxide pathway. In Texas in 2021 at least 19 people died of carbon monoxide from generators and cars run indoors.

Instead: keep the gas on, and install carbon monoxide alarms.

Do not rely on fixed warming centres alone

CONTRAINDICATED AS FRAMED SKEPTIC; CONFIRMED

In Buffalo, 17 people died on foot in snowbanks and 11 at home without heat; more than 420 emergency calls went unanswered. A centre nobody can reach saves no one.

* Suggested formulation: check with a health worker before sharing.

Instead: pair centres with outreach vans and phone lists for people living alone or on medical devices.

Do not buy heating-degree-day or weather cover

`CONTRAINDICATED` `ECONOMIST; CONFIRMED BY STATION DATA`

It would pay out in a cold winter, the scenario El Niño makes least likely, and nothing in the one-night-kills pattern. The real shock is a cash-timing gap.

Instead: a line of credit repaid from the federal award.

Do not route furnace or repair money through the city's ARPA home-repair program

`CONTRAINDICATED` `PHILANTHROPIST; CONFIRMED`

About $2.3M of $10M has been spent in two years, and lottery winners have waited two years. Money that cannot move in weeks saves no one this winter.

Do not announce a shutoff moratorium and call it done

`CONTRAINDICATED AS FRAMED` `SKEPTIC`

Deferred debt can turn into spring shutoffs. No Ohio source shows whether it does.

Instead: publish reconnection and arrears data every month through May.

Do not call the 400 HUD-funded beds "shelter beds"

`CONTRAINDICATED AS FRAMED` `VERIFICATION`

They are probably housing beds. Treating them as winter shelter capacity would misplan both the Code Blue overflow and the bridge reserve.

* Suggested formulation: check with a health worker before sharing.

WHAT HAS HAPPENED HERE BEFORE

THE RECORD

DATE	EVENT	TOLL	WHAT FOLLOWED
Nov 1913	Great Lakes storm: 22.2 in of snow, gusts to 79 mph	Power out for days	Five years moving utility cables underground
Nov 1950	Thanksgiving blizzard, 22.1 in	23 dead in the Cleveland area	National Guard called in
25–27 Jan 1978	Great Blizzard: Cleveland's record low pressure, 82 mph gusts	51 Ohio deaths of about 90; 110,000 Greater Cleveland homes without power	Ohio Turnpike closed end to end for the first time
1997–98	Very strong El Niño	Lake Erie ice 5.4%, the record low; lowest winter temperature 12 °F	—
Jan 2014	Polar vortex	Record −11 °F at Cleveland	Schools closed across Ohio
Aug 2014	Algal bloom over Toledo's intake	Half a million without drinkable water	SB 1 on fertiliser and manure, 2015
Dec 2022	Winter Storm Elliott: 3.6 in of snow in Cleveland	16 Ohio deaths: 8 in vehicles, 6 in one heating fire	—
Nov 2025	Federal LIHEAP arrives late; city pauses its own shutoffs	—	30-day moratorium from 7 Nov
Jan 2026	Cold spell	430+ visitors at four warming centres	—
Sep 2026	LIHEAP on a stopgap to 11 Dec	—	This briefing

The recurrence nobody has noticed, from the Historian: Cleveland's worst winter deaths came from sudden wind and cold inside otherwise ordinary winters. The forecast that makes a city relax is the one that precedes the killing night.

* Suggested formulation: check with a health worker before sharing.

WHERE THE EXPERTS DISAGREE — RESOLVED AND OPEN

- **El Niño makes this winter safer** versus **it makes no difference**. Settled in between by the station record: the average hazard falls, the tail does not, and mortality tracks neither closely.
- **60 extra shelter beds** versus **about 100 unsheltered people and existing levy-funded overflow**. Settled toward the verifier: 30 beds and one team.
- **A $0.9M reconnect fund** versus **Winter Crisis already pays reconnection**. Settled: held as a contingency.
- **Open:** whether Winter Crisis is funded for 2026–27, and when HUD announces awards. These two facts decide how much of the envelope is needed.

WHAT NOBODY COVERED — BLIND SPOT

No lens found the Cuyahoga medical examiner's hypothermia or carbon monoxide counts for recent winters; the site would not resolve. No lens found disconnection or arrears figures for Enbridge, FirstEnergy, or Cleveland Public Power, so the size of the shutoff problem is inferred, not measured. No lens reached East Cleveland, a separate and poorer city inside the same lake-effect belt. And no lens looked at the overdose crisis in the cold: overdose deaths are four times higher in the county's low-opportunity tracts, and nobody knows how many happen outdoors in winter.

HOW TO READ THIS — METHOD

Each briefing starts from a single, precise place, such as a village, a ward, or a district, because that is the scale at which useful actions differ. We first work out what can honestly be measured there. That usually means the Climate Impact Lab's published figures where they exist, the local rainfall record, census data, and the history of what has actually gone wrong in that place. Where no reliable number exists, we say so plainly rather than stretch a regional figure to fit.

We then put the same question to seven expert perspectives, each run as a separate AI research agent working from real sources. Each one asks what people here can do before the peak to keep each other alive.

- **The Emergency Manager** looks at what failed operationally in past events.
- **The Public Health Scientist** looks at who is most at risk and which interventions have measured effects.

* Suggested formulation: check with a health worker before sharing.

- **The Frontline Responder** looks at how the community actually works and who people trust.
- **The Skeptic** looks for advice that has backfired in comparable places.
- **The Economist** asks who pays, who carries the risk uninsured, and whether any money is pre-arranged to arrive on a trigger.
- **The Historian** looks for the pattern that keeps repeating.
- **The Philanthropist** asks where a grant would do the most good, and how much money the place could realistically absorb before the peak.

The seven briefs are then reconciled. Agreement between lenses counts for less than it appears to, because they often draw on the same famous studies. Where one lens recommends something another calls harmful, both sides are recorded rather than quietly resolved. Every surviving action needs an owner, a date, a cost, and a named payer, or it is marked unfunded. The funding recommendation becomes a bounded range: a floor, a target, and a ceiling, each with its arithmetic shown. Finally, a separate set of agents checks every recommendation against primary sources. They test whether the evidence is real, whether it came from somewhere like this place, and whether the action can be done here in time. Anything found to be harmful is kept in the report under its own heading, because advice that would cost lives is the most important thing a reader can be warned about. The result is a strong, source-checked hypothesis from a simulated panel, not a consensus of the field, and each briefing says so.

This run. The place was given as a city, not a ward, so actions target the east-side neighbourhoods the lenses named without ward-level verification. Because El Niño lowers rather than raises Cleveland's average hazard, the premise was tested directly against 76 winters of station data before any action was ranked. Web search was exhausted; all twelve agents worked by direct fetch. Five verifiers checked seventeen recommendations.

The El Niño projection covers *heat only* and publishes no US figure; this is a cold-season hazard it does not measure. The Adaptation Inventory holds three relevant US entries, all cited, and nothing on energy assistance or shutoff protection.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Cleveland, Ohio. Climate Impact Lab impact region, as the report records it: City of Cleveland, about 363,600 people, inside Cuyahoga County (several Climate Impact Lab regions). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February

* Suggested formulation: check with a health worker before sharing.

2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.

- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted; the closest documented precedent for each action, or its absence, is shown in that action's tags.
- **Health wording.** Passages where a wording error could hurt someone (unsafe heating, fire, generators, and carbon monoxide) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Gaps specific to this run are recorded in the verification banner and in each action's tags.

References (verification status in brackets) · Station record ACIS StnData, CLEthr · ONI CPC v6 · oni.ascii.txt · Lake Erie ice GLERL [computed in session] · El Niño impacts climate.gov [confirmed] · CPC discussion ensodisc · Energy burden ACEEE [corrected: national 8.3%] · LIHEAP funding liheapch.acf.gov · NEADA · NEADA 2026–27 costs [confirmed] · Gas bills Signal Cleveland [confirmed] · HB 173 veto Signal Cleveland [confirmed] ·

* Suggested formulation: check with a health worker before sharing.

Reconnect terms OCC [confirmed; 2026–27 order pending] · HEAP and Step Forward Signal Cleveland · stepforwardtoday.org [corrected] · Warming centres Signal Cleveland [confirmed] · LMM shelter lutheranmetro.org · Unsheltered plan Signal Cleveland · HUD beds Signal Cleveland · HUD rule changes [corrected: housing beds] · ARPA home repair Signal Cleveland [confirmed] · Housing Court kiosks clevelandhousingcourt.org · NEOCH neoch.org · Chirakijja, Jayachandran and Ong doi.org [confirmed] · Memmott 2023 PMC10025124 [confirmed; no mortality effect of its own] · Carley 2022 PMC9457251 [corrected] · Memmott 2024 PMC11538516 [confirmed] · Kinney 2015 doi.org [confirmed] · Howden-Chapman 2007 PMC1808149 · Lazo Green 2024 PMC11293810 [confirmed with caveat] · McMaster 2025 PMC12031699 · Bhullar 2026 doi.org [confirmed] · Przepyszny and Jenkins 2007, PMID 17721177 [found] · 1913 storm wikipedia · Blizzards Encyclopedia of Cleveland History · 1978 wikipedia [corrected] · Elliott wikipedia [confirmed; fire unlocated] · 2014 cold wikipedia [confirmed] · Toledo 2014 wikipedia · SB 1 legislature.ohio.gov [partially confirmed] · Texas 2021 wikipedia · Struck or unverified: Buffalo's toll of 47 (41 used); people freezing in stranded cars in 1978; El Niño snowfall figures from memory (replaced by station data); H2Ohio's founding; the Winter Reconnect Order's origin; Cleveland Public Power following the order voluntarily; Ohio's power to advance HEAP; Winter Crisis funding for 2026–27; the bulk-fuel threshold; the Red Cross program's CO alarm coverage; the $75 bed-night and $500 arrears unit costs.

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (queried via wp-json/wp/v2/interventions; US winter weather advisories entry; US contingent loans entry; US community health centres entry)

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